

Drug summary (high-level)

Margetuximab (MARGENZA, MGAH22) is a Fc-engineered anti-HER2 IgG1κ monoclonal antibody approved in the US for adults with metastatic HER2-positive breast cancer after ≥2 prior anti-HER2 regimens (≥1 in the metastatic setting), given in combination with chemotherapy. ([FDA Access Data](#))

Compared with trastuzumab, it binds the same HER2 epitope but has an engineered Fc region that increases affinity for activating FcγRIIIa (CD16A) and reduces affinity for inhibitory FcγRIIb (CD32B), thereby enhancing antibody-dependent cellular cytotoxicity (ADCC) and NK-cell/macrophage-mediated killing of HER2-positive tumor cells. ([PMC](#))

The pivotal phase III SOPHIA trial and subsequent analyses show modest but statistically significant PFS improvement vs trastuzumab plus chemotherapy, with exploratory OS benefit concentrated in patients carrying at least one low-affinity CD16A-158F allele, while no overall OS advantage was seen in the intention-to-treat population. ([JAMA Network](#))

Guidelines (NCCN 2024–2025; ESMO 2021–2024) position margetuximab+chemotherapy as a later-line HER2-directed option (typically 4th-line and beyond) after trastuzumab, pertuzumab, and an ADC (often T-DM1 and/or trastuzumab deruxtecan). ([Margenza](#))

2. Identifiers & synonyms

- **Generic name:** Margetuximab
 - **US brand name:** MARGENZA([FDA Access Data](#))
 - **Common research names:** MGAH22, margetuximab-cmkb([EMBL-EBI](#))
 - **DrugBank:** DB14967([DrugBank](#))
 - **ChEMBL compound ID:** ChEMBL2364649([EMBL-EBI](#))
 - **NCIt code:** C91733([DGIdb](#))
 - **ATC:** L01FD06 (HER2/neu receptor antagonist)([Wikipedia](#))
 - **Route:** Intravenous infusion([FDA Access Data](#))
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3. Molecular targets (human; HGNC level)

Primary pharmacologic target (tumor cell):

- **ERBB2 (HER2)** – receptor tyrosine kinase overexpressed/amplified in HER2+ breast cancer; margetuximab binds the HER2 extracellular domain at the same epitope as trastuzumab. ([Cancer.gov](#))

Key effector-cell receptors engaged via Fc:

- **FCGR3A (CD16A)** – activating FcγR on NK cells and some macrophages; Fc engineering increases affinity for both CD16A-158V and -158F allotypes to enhance ADCC. ([PMC](#))
- **FCGR2B (CD32B)** – inhibitory FcγR on B cells/macrophages; Fc engineering decreases affinity, reducing inhibitory signaling and further biasing toward immune activation. ([PMC](#))

These receptors are not “classical drug targets” in the sense of HER2 inhibition, but they are integral to the therapeutic pharmacodynamics (ADCC, phagocytosis, cytokine production).

4. Mechanism of action and key pathways

4.1 Core MoA

1. HER2 pathway blockade

- Binds HER2 and inhibits ligand-independent HER2 signaling, reducing downstream PI3K–AKT–mTOR and RAS–MAPK pathway activation and tumor cell proliferation, similar to trastuzumab. ([Oncolink](#))
- Contributes to **downregulation of ERBB2 signaling**, including receptor internalization/degradation and reduced shedding of the HER2 extracellular domain. ([Margenza](#))

2. Fc-enhanced immune effector recruitment

- Engineered Fc domain increases binding to **CD16A** and reduces binding to **CD32B**, resulting in stronger activation of NK cells and macrophages compared with trastuzumab. ([PMC](#))
- Preclinical and early-phase ex vivo studies show higher ADCC potency and greater NK-cell activation/expansion with margetuximab vs trastuzumab across all CD16A genotypes. ([PMC](#))

3. Immune-modulatory sequelae

- Fc-mediated tumor-targeting promotes **NK-cell cytotoxicity, cytokine release (e.g., IFN-γ)**, and **FcγR-dependent phagocytosis** by macrophages. ([Frontiers](#))

- Fc-dependent immune activation can upregulate PD-L1 and other interferon-stimulated genes in the tumor microenvironment, forming a rationale for combinations with checkpoint inhibitors. ([Macrogenics](#))

4.2 Pathway sets most directly implicated (with canonical IDs)

HER2/ERBB2 signaling & downregulation

1. **ErbB signaling pathway – KEGG hsa04012**
– Canonical HER family signaling (HER2 homo-/heterodimers → RAS–RAF–MEK–ERK, PI3K–AKT). ([KEGG](#))
2. **ERBB2 signaling in cancer – Reactome R-HSA-1227990 / MSigDB**
REACTOME_ERBB2_SIGNALING ([Reactome](#))
3. **Downregulation of ERBB2 signaling – Reactome R-HSA-8863795 / MSigDB**
REACTOME_DOWNREGULATION_OF_ERBB2_SIGNALING
– Internalization, ubiquitination, and degradation of HER2 and heterodimers; relevant to antibody-mediated down-modulation. ([Reactome](#))
4. **HALLMARK_PI3K_AKT_MTOR_SIGNALING** (MSigDB) – down-modulated indirectly by effective HER2 blockade. ([CEBS](#))

FcγR, NK-cell, and innate immune pathways

5. **Natural killer cell mediated cytotoxicity – KEGG hsa04650 / MSigDB**
KEGG_NATURAL_KILLER_CELL_MEDIATED_CYTOTOXICITY
– Pathway engaged via CD16A-mediated NK activation; explicitly referenced as central to ADCC by Fc-engineered antibodies. ([KEGG](#))
6. **Fc gamma R-mediated phagocytosis – KEGG hsa04666 / MSigDB**
KEGG_FC_GAMMA_R_MEDIATED_PHAGOCYTOSIS
– Macrophage/neutrophil uptake of antibody-opsonized cells via FcγRs; mechanistically relevant for margetuximab-coated tumor cells. ([Genome](#))
7. **Fcγ receptor (FCGR)–dependent phagocytosis – Reactome R-HSA-2029480 / MSigDB**
REACTOME_FCGAMMA_RECEPTOR_FCGR_DEPENDENT_PHAGOCYTOSIS ([Reactome](#))
8. **HALLMARK_INTERFERON_GAMMA_RESPONSE** – genes induced by IFN-γ, implicated by increased IFN-γ secretion and interferon-related signaling in Fc-engineered antibody models and margetuximab/checkpoint combinations. ([GSEA MSigDB](#))

9. **HALLMARK_INFLAMMATORY_RESPONSE** – broad inflammatory cytokine and myeloid activation signature that often accompanies Fc-mediated effector activity. ([GSEA MSigDB](#))
10. **HALLMARK_COMPLEMENT** – complement activation is not the primary mechanism but can accompany IgG1 tumor-bound antibodies and may contribute to cytotoxicity and inflammation in some settings. ([CEBS](#))

Because there are essentially **no published, large-scale transcriptomic or “omics” datasets specifically for margetuximab in breast cancer**, most of the pathway mapping above is mechanistic (target→pathway) rather than driven by GSEA on drug-treated tumors. Where inference is used, it is explicitly anchored to curated pathway definitions and experimental MoA data, not guessed.

5. Sensitivity / predictive biology

5.1 CD16A (FCGR3A) polymorphisms

- The SOPHIA trial’s exploratory biomarker analysis and later JCO “final OS” report show that **CD16A-158F carriers** (FF or FV) derive greater relative benefit from margetuximab vs trastuzumab:
 - **PFS:** Presence of a CD16A-158F allele predicted improved PFS with margetuximab over trastuzumab. ([JAMA Network](#))
 - **OS:** Final OS analyses showed **no significant ITT OS difference**, but suggested improved OS for margetuximab in **CD16A-158FF** patients (HR ≈0.72; 95% CI 0.52–1.00), whereas **CD16A-158VV** patients appeared to fare better with trastuzumab (HR ≈1.77; 95% CI 1.01–3.12). ([PubMed](#))
- Mechanistic basis: Fc engineering increases affinity for both CD16A allotypes but yields proportionally greater improvement for the low-affinity 158F allele, enhancing ADCC in patients who derive less benefit from trastuzumab. ([PMC](#))

Pathways implicated in sensitivity:

- **FCGR3A/CD16A signaling + NK-cell cytotoxicity** (KEGG hsa04650; FcγR-dependent pathways listed above). Patients in whom these pathways are functionally weaker with trastuzumab (CD16A-158F) become “rescued” by Fc-engineering. ([Frontiers](#))

5.2 HER2 expression & activation

- All pivotal trials restrict to IHC 3+ or ISH-amplified **HER2-positive** tumors; high HER2 expression remains a prerequisite.

- No HER2-low (IHC 1+/2+ non-amplified) breast cancer cohorts with outcome data for margetuximab have been published to date; HER2-low ADCs (e.g., trastuzumab deruxtecan) dominate that space instead. ([DrugBank](#))

Thus, HER2 pathway activity (ErbB signaling) is a **necessary but not discriminative** biomarker within HER2-positive disease for margetuximab, as for trastuzumab.

5.3 Tumor microenvironment / immune contexture (indirect)

- Reviews of margetuximab and Fc-engineered antibodies emphasize the importance of **pre-existing or inducible NK-cell and myeloid effector function**, including ADCC capacity and FcγR expression, for optimal benefit. ([PMC](#))
- In vitro/ex vivo studies demonstrate increased NK-cell activation markers and cytokine production (e.g., IFN-γ) with margetuximab vs trastuzumab, supporting links to **HALLMARK_INTERFERON_GAMMA_RESPONSE** and **HALLMARK_INFLAMMATORY_RESPONSE**, although clinical correlations in breast cancer patients have not been rigorously quantified. ([PMC](#))

Bottom line (sensitivity):

The **only robust, prospectively explored predictive factor** is **CD16A genotype**, with stronger relative benefit in CD16A-158F carriers. HER2 positivity is required but not uniquely predictive, and “immune signature” sensitivity remains conceptual rather than validated.

6. Resistance and escape biology

There are **no dedicated resistance-omics datasets specific to margetuximab** in breast cancer. Mechanistically, resistance is expected to overlap heavily with other HER2 antibodies, but per your constraints, I’ll only highlight factors with explicit links either to margetuximab or to the FcγR biology it exploits.

6.1 Lack of global OS advantage and late-line setting

- Despite improved PFS, final OS in SOPHIA was not significantly different between arms, suggesting that **late-line HER2-positive metastatic disease is dominated by multi-factorial resistance** (clonal heterogeneity, prior ADCs/TKIs, aggressive biology) that Fc-engineering alone cannot fully overcome. ([PubMed](#))

This points to resistance pathways *upstream* (loss of HER2 dependence) and *downstream* (hyperactivated PI3K/AKT/MAPK) rather than to a specific failure of FcγR biology.

6.2 FcγR-axis limitations

- The same CD16A genotypes that favor trastuzumab (158VV) appear to **limit relative margetuximab benefit**, possibly because baseline ADCC is already high and Fc-engineering adds little or even biases receptor usage unfavorably. ([PubMed](#))

Here, “resistance” is more relative (vs trastuzumab) than absolute – but it highlights how **FcγR pathway polymorphisms** (FCGR3A, FCGR2A/B) and their downstream signaling (NK/T-cell exhaustion, inhibitory myeloid polarization) can constrain efficacy. ([Journal of Infection and Control](#))

6.3 Classic HER2-pathway resistance (class-level)

Direct margetuximab-specific data are sparse, but reviews note that classical HER2-targeted resistance mechanisms (p95HER2 truncation, PTEN loss, PI3K pathway activation, HER2 cross-talk with other RTKs) are very likely relevant to margetuximab, as they reduce dependency on extracellular HER2 signaling. ([ResearchGate](#))

Because these have not been formally dissected in margetuximab cohorts, I will **not** list specific gene-level resistance signatures for margetuximab, but mechanistically they map to:

- **PI3K–AKT–mTOR signaling (HALLMARK_PI3K_AKT_MTOR_SIGNALING)** ([CEBS](#))
- **RAS–MAPK cascade** (within ErbB signaling). ([KEGG](#))

7. Clinical evidence in human breast cancer (2019–2025 focus)

7.1 Metastatic HER2-positive breast cancer (all HR subtypes)

SOPHIA (NCT02492711) – Phase III, global

- **Population:** 536 patients with HER2-positive advanced/metastatic breast cancer, previously treated with ≥2 anti-HER2 regimens (≥1 metastatic); mixed HR-positive and HR-negative. ([JAMA Network](#))
- **Design:** Margetuximab 15 mg/kg q3w + investigator-choice chemo vs trastuzumab + chemo (capecitabine, eribulin, gemcitabine, or vinorelbine). ([JAMA Network](#))
- **Primary endpoint (PFS):**
 - Interim analysis: mPFS 5.8 vs 4.9 months; HR 0.76 (95% CI 0.59–0.98); 24% relative risk reduction. ([JAMA Network](#))
- **Final OS (JCO 2023 + press release):**

- ITT OS: **21.6 vs 21.9 months**; no statistically significant difference. ([PubMed](#))
- CD16A-158FF subgroup: OS 23.6 vs 19.2 months; HR 0.72 (95% CI 0.52–1.00), suggesting a possible OS benefit but not powered for definitive proof. ([PubMed](#))
- **HR+/HR-:** Published reports describe similar relative benefit across HR strata; there are no strong signals that HR status dramatically modifies margetuximab effect, but data are not granular enough to claim subtype-specific predictive value. ([Annals of Oncology](#))

Chinese Phase II bridging study (NCT04262804)

- **Design:** Open-label, randomized phase II bridging in pretreated HER2-positive advanced/metastatic BC (Chinese population), margetuximab+chemo vs trastuzumab+chemo, paralleling SOPHIA. ([PMC](#))
- **Results:**
 - mPFS 5.8 vs 4.9 months (HR 0.69; 95% CI 0.46–1.04); ORR also numerically higher; safety similar to SOPHIA. ([PMC](#))
 - CD16A F-allele carriers again had numerically better outcomes with margetuximab than trastuzumab, consistent with SOPHIA. ([Translational Breast Cancer Research](#))

Real-world case series and reports (late-line)

- **Frontiers in Oncology 2024 case series (HER2+ mBC; margetuximab in routine practice)** – describes later-line margetuximab+chemo in women heavily pre-treated with anti-HER2 therapies (including ADCs), suggesting disease control and occasional responses, largely aligning with SOPHIA expectations but with small numbers. ([Frontiers](#))
- **Chinese case report (Anti-Cancer Drugs 2023):** dramatic response in a postmenopausal HER2+/HR- patient with liver metastases after prior trastuzumab/pertuzumab and pyrotinib, achieving complete response of liver lesions on margetuximab+chemo. ([PMC](#))

Economic analysis (US/China)

- A 2022 cost-effectiveness model concluded that **at current pricing, margetuximab+chemo is not cost-effective** vs trastuzumab+chemo in pretreated HER2+ ABC in both US and China, though price reductions could change this. ([Frontiers](#))

7.2 Early-stage HER2-positive breast cancer

- As of 2024–2025, **no approved early-stage indication**.
- **MARGOT (NCT04425018)**: Phase II neoadjuvant trial comparing paclitaxel/margetuximab/pertuzumab (TMP) vs paclitaxel/trastuzumab/pertuzumab (THP) in stage II–III HER2-positive breast cancer; endpoints include pCR and immune/biomarker analyses. Results not yet mature. ([Clinical Trials](#))

7.3 Other breast subtypes

- **HER2-low BC**: No published dedicated margetuximab trials; practice is dominated by trastuzumab deruxtecan and other ADCs. ([Verywell Health](#))
- **Triple-negative BC (TNBC)**: Margetuximab has appeared in **early-phase basket trials** that included TNBC and other tumors, but without clear breast-specific efficacy reporting; thus no robust TNBC-specific conclusion can be drawn. ([DrugBank](#))

So, across “all subtypes,” the **only well-defined clinical niche is HER2-positive disease**, with HR+ and HR– treated similarly; HER2-low and HER2-negative subtypes currently lack meaningful margetuximab data.

8. Dosing, combinations, and clinical positioning

8.1 Dosing (label)

- **Dose**: 15 mg/kg IV every 3 weeks (21-day cycle) until progression or unacceptable toxicity. ([FDA Access Data](#))
- **Infusion**:
 - First dose over 120 min; subsequent doses over ≥30 min, every 3 weeks. ([FDA Access Data](#))
- **Sequence with chemotherapy**: On days when both are given, chemo first, then margetuximab. ([FDA Access Data](#))

8.2 Combination partners

In SOPHIA and the label, margetuximab is combined with **single-agent chemotherapy**, typically:

- Capecitabine
- Eribulin
- Gemcitabine

- Vinorelbine([JAMA Network](#))

Guidelines echo these partners as acceptable; no clear superiority of one chemotherapy backbone over another has emerged.([ASCO Publications](#))

8.3 Placement in modern HER2+ metastatic sequences (HR+ and HR-)

Contemporary NCCN/ESMO algorithms typically:

1. **1st line:** Taxane + trastuzumab + pertuzumab (THP) or equivalent.
2. **2nd line:** Trastuzumab deruxtecan (T-DXd) in many jurisdictions.
3. **3rd line:** Tucatinib + trastuzumab + capecitabine (HER2CLIMB), or other options.
4. **4th-line and beyond:** Margetuximab+chemo is listed as an option, especially in the US (NCCN Category 2A; late-line HER2+ mBC).(Margenza)

This applies similarly to **HR+ and HR- HER2+** disease; the HR status mainly affects endocrine use and co-targeting, not margetuximab's place per se.

9. Safety, contraindications, and practical issues

9.1 Contraindications

- **Formal contraindications:** None listed in the FDA label.([FDA Access Data](#))

9.2 Major warnings (class-consistent)

From the US prescribing information:([FDA Access Data](#))

1. **Left ventricular dysfunction (boxed warning)**
 - Cardiac dysfunction (drop in LVEF) occurred in ~1.9% in SOPHIA.
 - Requires baseline and q3-month LVEF monitoring; dosing interruptions and discontinuation criteria are specified (e.g., ≥16% absolute LVEF drop, or LVEF < LLN/50% with ≥10% drop).
2. **Embryo-fetal toxicity (boxed warning)**
 - Animal data and HER2-antibody class experience indicate risk of oligohydramnios, fetal renal/pulmonary anomalies, and neonatal death; pregnancy must be avoided during treatment and for 4 months after last dose.
3. **Infusion-related reactions (IRRs)**
 - Occurred in ~13% of patients; Grade 3 in ~1.5%; usually cycle 1.

- Management: slow or interrupt infusion; consider premedication (antihistamines, corticosteroids, antipyretics); permanently discontinue for severe or life-threatening IRRs.

Common adverse reactions (>10%, mostly overlapping with chemo):

fatigue/asthenia, nausea, diarrhea, vomiting, constipation, headache, fever, alopecia, abdominal pain, peripheral neuropathy, arthralgia/myalgia, cough, decreased appetite, dyspnea, IRRs, PPE, extremity pain. ([FDA Access Data](#))

9.3 Practical HER2-BC caveats

- Use caution in patients with significant pre-existing cardiac disease or baseline LVEF <50%; these populations were largely excluded from SOPHIA. ([FDA Access Data](#))
- Given modest PFS gain and no ITT OS advantage, many clinicians reserve margetuximab for patients whose **CD16A genotype suggests benefit (158F carriers)** or where other standard late-line options are exhausted/unavailable; this is reflected in expert reviews rather than mandated in guidelines. ([Wiley Online Library](#))

10. Evidence across breast cancer subtypes (treated “equally” in discussion)

- **HER2+/HR+ and HER2+/HR– (all-comers HER2+):**
 - Direct trial evidence (SOPHIA, Chinese phase II) includes both HR+ and HR–; efficacy analyses do not show major divergence by HR status. Margetuximab is therefore used in both HR+ and HR– HER2+ metastatic patients after ≥2 prior anti-HER2 regimens, regardless of hormone receptor status. ([JAMA Network](#))
- **HER2-low (HR+ or HR–):**
 - No clinically meaningful margetuximab data; standard is trastuzumab deruxtecan and other ADCs. ([Verywell Health](#))
- **Triple-negative (HER2-negative) BC:**
 - Margetuximab has appeared in basket trials including TNBC, but without subtype-specific breast outcomes; there is **no current rationale** to use it in HER2-negative TNBC outside a trial. ([DrugBank](#))

So, “equally” here mostly means being clear that **only HER2-positive subtypes of breast cancer have robust evidence; the others effectively have none at present.**

11. FINAL OUTPUT SECTION — Pathway evidence table

Below is a structured summary of the **key pathways** linked to margetuximab in human breast cancer, what role they play, and the type/strength of evidence.

Pathway / Gene Set	ID / Source	Role (MoA / Sensitivity / Resistance / Subtype context)	Direction / Modulation in context of margetuximab*	Evidence basis (breast-cancer-relevant)	Key references
ErbB signaling	KEGG hsa04012	Core MoA pathway downstream of HER2	Down-regulated when HER2 is blocked; contributes to anti-proliferative effect	HER2 is primary target; HER2-antibody blockade reduces downstream PI3K/MAPK signaling in HER2+ BC	(KEGG)
ERBB2 signaling in cancer	Reactome R-HSA-1227990; MSigDB REACTOME_ERBB2_SIGNALING	HER2-driven oncogenic signaling	Inhibited by HER2 binding and internalization	HER2+ BC depends on ERBB2 pathway; margetuximab shares epitope and	(Reactome)

Pathway / Gene Set	ID / Source	Role (MoA / Sensitivity / Resistance / Subtype context)	Direction / Modulation in context of margetuximab*	Evidence basis (breast-cancer-relevant)	Key references
Downregulation of ERBB2 signaling	Reactome R-HSA-8863795; MSigDB REACTOME_DOWNREGULATION_OF_ERBB2_SIGNALING	Canonical HER2 down-modulation pathway	Activated (more downregulation) via antibody-mediated internalization/degradation	antiproliferative effect with trastuzumab Reactome describes ubiquitin-dependent downregulation; margetuximab reduces HER2 ECD shedding and signaling	(Reactome)
PI3K-AKT-mTOR	HALLMARK_PI3K_AKT_MTOR_SIGNALING (MSigDB)	Downstream HER2 survival/pr	Functionally down-regulated with effective	PI3K/AKT a known mediator	(CEBS)

Pathway / Gene Set	ID / Source	Role (MoA / Sensitivity / Resistance / Subtype context)	Direction / Modulation in context of margetuximab*	Evidence basis (breast-cancer-relevant)	Key references
signaling		proliferation axis	HER2 blockade; hyperactivation likely mediates resistance	role of HER2-targeted resistance; implicated for class and thus relevant to margetuximab	
Natural killer cell mediated cytotoxicity	KEGG hsa04650; MSigDB KEGG_NATURAL_KILLER_CELL_MEDIATED_CYTOTOXICITY	Central MoA for Fc-engineered ADCC; sensitivity axis	Up-activated vs trastuzumab via stronger CD16A engagement	greater NK activation and ADCC with margetuximab; CD16A genotype modifie	(PMC)

Pathway / Gene Set	ID / Source	Role (MoA / Sensitivity / Resistance / Subtype context)	Direction / Modulation in context of margetuximab*	Evidence basis Key (breast-cancer-relevance)
FcγR-mediated phagocytosis	KEGG hsa04666; MSigDB KEGG_FC_GAMMA_R_MEDIATED_PHAGOCYTOSIS	Macrophage-mediated clearance of opsonized tumor cells (MoA)	Up-activated when tumor cells are opsonized; enhanced by Fc engineering favoring activating FcγRs	KEGG/Reactome define pathway; FcγR biology and Fc-engineered antibodies (including margetuximab) shown to enhance FcγR-dependent functions (Genome)
FcγR-dependent	Reactome R-HSA-2029480; MSigDB REACTOME_FCGAMMA_RECEPTOR_FCGR_DEPENDENT_PHAGOCYTOSIS	As above, with more mechanistic	Up-activated via IgG1-FcγR clustering	Reactome pathway; FcγR- (Reactome)

Pathway / Gene Set	ID / Source	Role (MoA / Sensitivity / Resistance / Subtype context)	Direction / Modulation in context of margetuximab*	Evidence basis Key (breast-cancer-relevance)
phagocytosis		c granularity		optimized mAb (margetuximab) highlighted as example in FcγR reviews
IFN-γ response	HALLMARK_INTERFERON_GAMMA_RESPONSE	Reflects NK/T-cell cytokine signaling; relevant for immune activation and potential synergy with checkpoint blockade	Likely up-regulated in responsive tumors due to enhanced NK activity; some Fc-engineered antibody models show increased IFN-γ	Gene set defined by MSigDB ; combination trials (margetuximab + PD-1 inhibitors) and Fc-engineering studies show increased IFN-

([GSEA](#)
[MSigDB](#))

Pathway / Gene Set	ID / Source	Role (MoA / Sensitivity / Resistance / Subtype context)	Direction / Modulation in context of margetuximab*	Evidence basis (breast-cancer-relevant)	Key references
Inflammatory response	HALLMARK_INFLAMMATORY_RESPONSE	Broad cytokine/myeloid activation; relates to both efficacy and toxicity (IRRs, cytokine release)	Up-regulated acutely during strong FcγR engagement; chronic activation may contribute to immune exhaustion	γ-linked immune activation Hallmark definition plus NK-/ADCC-focused reviews of Fc-engineered antibodies and clinical IRR patterns with margetuximab	(GSEA A MSigDB)
HER2+/FcγR genotype interaction	Not a single pathway; involves FCGR3A/2A/B loci and NK-cell signaling	Sensitivity/resistance axis defined by CD16A-158F/V; shapes	For CD16A-158F carriers, Fc engineering overcomes weaker baseline	Prospective CD16A analysis in SOPHIA and	(JAMA A Network)

Pathway / Gene Set	ID / Source	Role (MoA / Sensitivity / Resistance / Subtype context)		Evidence basis (breast-cancer-relevant)	Key references
		Direction / Modulation in context of margetuximab*			
		balance between activating and inhibitory FcγR signals	ADCC , improving PFS/OS vs trastuzumab; CD16A-VV may not benefit	Chinese phase II; FcγR mechanistic work	

*“Direction/modulation” reflects mechanistically expected changes compared with baseline or trastuzumab, not measured GSEA in patient tumors (no such datasets currently published for margetuximab in breast cancer).

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